

- **LLM Inference Engine**

A from-scratch LLM inference engine in PyTorch (KV cache, batching, paged memory)

[Github](#)

- Built a production-inspired LLM inference engine from scratch in PyTorch with GPT-2 and Mistral model paths, KV caching, continuous batching, paged memory, and asynchronous serving.
- Achieved a 2.22x decode-throughput speedup at 1,000 generated tokens and reduced CUDA time by 46.6% using a preallocated KV cache on an A100.
- Scaled GPT-2 124M throughput from 155 to 18,346 tokens/second through batched inference, a 118.3x gain at batch size 512 on an A100.
- Reduced peak KV-cache memory by up to 35.9% and delivered up to 3.2x higher throughput under 64 concurrent requests; covered the stack with 122 tests across 15 files.